1. **Which of the following best describes the design principle behind parameterizing a model architecture for use with tools like Optuna?** 1 point

- [x] It supports flexible experimentation by testing multiple architectural designs.
- [ ] It ensures Optuna trials are deterministic by locking random seeds.
- [ ] It simplifies the forward pass logic in PyTorch's *nn.Module*.
- [ ] It enforces model reproducibility through fixed layer structures.

2. **True or False: Both dropout rate and learning rate are regularization hyperparameters.** 1 point

- [x] False.
- [ ] True.

3. **You are defining a flexible CNN architecture with Optuna. Complete the code snippet below to allow for dynamic dropout selection for any value between 0.1 and 0.5.** 1 point

```
def objective(trial):
    n_layers = trial.suggest_int("n_layers", 1, 3)
    fc_size = trial.suggest_int("fc_size", 64, 512, step=64)

    # What goes here?
    __________________________

    model = FlexibleCNN(n_layers=n_layers, fc_size=fc_size, dropout_rate=dropout_rate)

```

Which line correctly completes the block?

- [ ]
```
dropout_rate = FlexibleCNN.suggest("dropout_rate", 0.1, 0.5)
```
- [ ]
```
dropout_rate = trial.suggest_categorical("dropout_rate", [0.1, 0.2, 0.3, 0.4, 0.5])
```
- [ ]
```
dropout_rate = trial.sample_float("dropout_rate", 0.1, 0.5)
```
- [x]
```
dropout_rate = trial.suggest_float("dropout_rate", 0.1, 0.5)
```

4. **Which of the following is a best practice when designing an optimization pipeline using Optuna?** 1 point

- [ ] Include only training hyperparameters like learning rate and batch size, not architectural ones.
- [x] Structure the search space to reflect dependencies between architectural components (e.g., kernel sizes only if a layer exists).
- [ ] Define all hyperparameters regardless of their dependencies (e.g. define kernel size for layers that may not exist)
- [ ] Test all possible combinations of hyperparameters

5. **Why might you prioritize evaluating model size and inference time in addition to accuracy?** 1 point

- [ ] Accuracy always guarantees generalization.
- [ ] To better understand the GPU temperature behavior.
- [ ] These values are easier to measure than accuracy.
- [x] To ensure your model meets real-world constraints like speed and memory.

6. **Consider this snippet from a flexible CNN:** 1 point

```
def forward(self, x):
    for layer in self.features:
        x = layer(x)

    x = torch.flatten(x, 1)
    if self.classifier is None:
        self._init_classifier(x.size(1))

    return self.classifier(x)
```

Why is the classifier initialized inside the forward method?

- [ ] To prevent the classifier from being saved in the model checkpoint.
- [ ] To delay initialization until the model is trained.
- [x] Because *x.size(1)* gives the dynamic flattened feature size, which depends on earlier layers.

- To reset the classifier weights every time.

7. **What insight can you gain by analyzing parameter importances after an Optuna optimization run?** 1 point

- You can identify which training samples contributed most to performance variation.
- You can identify which hyperparameters consistently led to underfitting.
- You can determine which combination of hyperparameter values optimizes the objective function
- You can determine which hyperparameters had the most influence on the optimization objective and prioritize them in future searches.

8. **How does the parallel coordinate plot help in understanding the results of a hyperparameter search?** 1 point

- It shows epoch-wise performance trends.
- It ranks models based on final test accuracy.
- It plots validation loss across models.
- It highlights optimal combinations of hyperparameter values using color intensity.

9. **In a hyperparameter tuning process, what is the primary role of a baseline model?** 1 point

- To confirm that your model is generalizing well to new data.
- To serve as a reference point against which performance improvements can be measured.
- To maximize model complexity during early experimentation
- To automate model selection based on predefined criteria.

10. **You aim to select the model that maximizes the following composite score:** 1 point

Score = 0.5 × Normalized Accuracy + 0.3 × Normalized Memory Size + 0.2 × Normalized Inference Time

Here are your top four models:

Model	Normalized Accuracy	Normalized Memory Size	Normalized Inference Time
A	0.8	0.7	0.3
B	0.6	0.7	0.9
C	0.7	0.6	0.95
D	0.75	0.8	0.2

Which model should you choose?

- D
- B
- A
- C